



Database Management System Lab

Open Ended Lab Report

CSE 2302

Rare Blood Emergency and Matching System (RBEMS)

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1. Problem Requirements with Clarification

Project Overview

The Rare Blood Emergency and Matching System (RBEMS) is a web-based database application designed to coordinate blood donation, storage, and emergency distribution across a small network of hospitals. Locating compatible blood — particularly for patients with rare phenotypes such as Rh-null, Bombay phenotype, or Kell-positive profiles — is time-critical and error-prone when handled manually with paper registers or disconnected spreadsheets. RBEMS centralizes donor records, live inventory, hospital requests, and inter-hospital transfers in a single relational database, and automates the rules that are normally applied by hand: donor eligibility, unit shelf-life, compatibility matching, and request prioritization.

Problem Statement

Blood banks and hospitals frequently need to answer four questions quickly: (1) Which donors are currently eligible to donate? (2) Which stored units, or which donors, are compatible with a specific and possibly rare recipient profile? (3) Which units are about to expire and must be used or discarded? and (4) Which hospital requests are most urgent and how should scarce units be allocated between hospitals? A manual process cannot reliably answer these questions under time pressure, especially when rare phenotypes are involved and the pool of compatible donors or units is very small. The system must therefore encode these rules directly in the database (through triggers, views, and stored procedures) rather than leaving them to human judgement alone, so that the correct answer is always computed consistently and instantly.

Objectives

- Maintain a reliable, centralized record of donors, patients, hospitals, inventory, and requests.
- Automatically determine donor eligibility using a donation-interval rule (90 days for male donors, 120 days for female/other donors).
- Automatically compute the expiry date and shelf-life status of every inventory unit at the moment it is entered (42 days for fresh whole blood, 10 years for cryopreserved units), and flag units expiring within 7 days.
- Provide a rare-blood compatibility cross-matching engine that layers Rh-null, Bombayphenotype, and Kell overrides on top of standard ABO/Rh compatibility rules.
- Allow authorized hospitals to submit blood requests that are automatically queued by priority (Emergency > Urgent > Routine).
- Give administrators a single dashboard to approve or reject requests and donor registrations, and to create and track inter-hospital unit transfers.
- Automatically log notifications when a request is submitted or a transfer is completed, so all stakeholders can track status without manual follow-up.

- Enforce basic security through role-based logins (Admin, Donor, Patient) and parameterized queries to prevent SQL injection.

Actors / Stakeholders

Actor	Description	Key Interactions
Administrator	System-wide operator seeded at setup (username: admin1).	Approve/reject hospital requests and donor registrations, create and complete inter-hospital transfers, delete donor/patient accounts, view all data (admin_login.php, admin_dashboard.php).
Donor	A registered person willing to donate blood; account requires admin approval before login is possible.	Self-register with phenotype profile, browse/search hospitals, confirm a donation appointment with a chosen patient and hospital (donor_register.php, donor_dashboard.php).
Patient	A registered person who needs blood/donor matching.	Self-register with blood-group profile, log in, view own profile and the list of approved donors and confirmed donation meetings (register.php, patient_dashboard.php).
Hospital	Represented as a reference entity (not a login role in the current build) on whose behalf requests and inventory are recorded.	Requests are submitted through hospital_request.php and inventory is attributed to a hospital in inventory_add.php / inventory.php.

Functional Requirements

The system implements eight core features, each clarified below with the assumption or business rule applied and the file(s) that realize it.

Donor Registration and Phenotype Profile

A visitor can create a donor account (register.php) or be registered directly (donor_register.php) by capturing identity, contact details, and a full phenotype profile: ABO group, Rh type (including Rh-null), Kell status, and a Bombay-phenotype flag. Clarification: donor accounts created through self-registration are inactive until an administrator approves them (is_approved = 0), so an unapproved donor cannot log in even with correct credentials.

Automatic Donation Eligibility

Eligibility is not stored as a flag that a human sets; it is derived automatically from the date of the donor's last donation using an interval rule of 90 days for male donors and 120 days for female/other donors. This is exposed to every page through the v_eligible_donors database view and mirrored in

functions.php::checkEligibility() so PHP-side matching logic (e.g., in compatibility_check.php) stays consistent with the database-side view.

Rare Blood Compatibility Cross-Matching

A recipient's ABO group, Rh type, Kell status, and Bombay-phenotype flag are submitted through compatibility_check.php, which calls the sp_find_compatible_inventory stored procedure to find compatible inventory units and also filters the eligible-donor view in PHP with isCompatible() to surface compatible donors as a fallback when no unit is in stock. Clarification/assumption stated in the README: Bombay-phenotype recipients can only receive Bombay-phenotype blood; Rh-null recipients can only receive Rh-null blood; Kell-negative recipients cannot receive Kell-positive blood. These overrides sit on top of standard ABO/Rh compatibility.

Inventory Shelf-Life and Expiry Alerts

When a unit is added (inventory_add.php), only the collection date and storage type are entered by the user; a database trigger (trg_inventory_expiry) automatically computes the expiry date — 42 days later for fresh whole blood, 10 years later for cryopreserved units. inventory.php lists every unit ordered by status and expiry, and flags any available unit with 7 or fewer days remaining. The v_expiring_soon view surfaces this same list on the main dashboard.

Hospital Request Submission with Priority Queueing

hospital_request.php lets a hospital submit a need (blood group, Rh type, Kell status, Bombay flag, number of units, priority, and a needed-by time). Clarification: priority is a simple three-level enumeration (Emergency, Urgent, Routine) and every query that lists requests orders them with FIELD(priority,'Emergency','Urgent','Routine') so the queue is always priority-first, then chronological — the ordering is enforced at query time rather than trusted to insertion order.

Administrator Approval and Inter-Hospital Transfers

admin_dashboard.php is the operational hub: it lists every Pending/Approved request and lets the admin approve or reject it; lists donor accounts awaiting approval and lets the admin approve or reject/delete them; and provides a transfer-creation form that links an approved request to either a specific available inventory unit (which is then marked Reserved) or, if no unit is specified, implies a donor will be mobilized instead. Open transfers can be marked Completed from the same screen.

Request Status Tracking and Notifications

request_list.php shows every request (all statuses) ordered first by status and then by priority, so hospitals can track progress end-to-end. notifications.php shows a running log that is populated automatically by database triggers (trg_request_notification, trg_transfer_completed) whenever a request is submitted or a transfer completes — no page has to manually insert a notification row.

Role-Based Access and Session Security

admin_login.php authenticates three separate roles — patient, donor, and admin — against three separate tables, regenerates the session ID on every successful login, and clears any admin session data

when a patient/donor logs in (and vice versa) to prevent privilege leakage between roles. `requireLogin()`, `requireDonorLogin()`, and `requirePatientLogin()` guard the corresponding dashboards so an unauthenticated visitor is redirected before any protected data loads.

Non-Functional Requirements

- Data integrity: all write operations use PDO prepared statements with bound parameters, and business rules that must never be bypassed (expiry calculation, notification logging) are enforced in the database itself via triggers rather than only in PHP.
- Security: passwords are hashed (`password_hash` / `password_verify` for donor and patient accounts; hashed comparison for admin), and session state is regenerated on login to reduce session-fixation risk.
- Usability: every page follows a consistent card-based layout with a shared header/footer, colour-coded status badges (e.g., priority and status), and inline success/error banners after form submission.
- Auditability: notifications and transfer records provide a basic audit trail of who requested what and when it was fulfilled.
- Portability: the system runs on any standard PHP + MySQL stack (e.g., XAMPP/WAMP) with no external service dependencies.

Assumptions and Business Rules

The following assumptions were made explicit so that grading and future maintenance are unambiguous, matching the notes documented in the project README:

- Donation interval: 90 days for male donors, 120 days for female/other donors — a simplified standard blood-bank rule.
- Shelf life: 42 days for fresh whole blood, 10 years for cryopreserved units.
- Compatibility overrides: Bombay-phenotype recipients → Bombay-phenotype blood only; Rhnull recipients → Rh-null blood only; Kell-negative recipients cannot receive Kell-positive blood.
- Usernames for patient/donor self-registration are restricted to 3–50 letters (validated both client-side and server-side with a regular expression) to keep the demo dataset simple; the admin account is exempt from this pattern.
- A single seeded administrator account (`admin1`) is provided for grading/demo purposes rather than a full admin-management workflow.

Scope and Limitations

The current build focuses on the core data-management and rule-automation problem rather than production concerns: there is no SMS/e-mail delivery of notifications (they are logged in-database only), hospitals do not have their own login role in this version (requests are attributed to a hospital chosen from a list), and the system assumes a single-institution deployment rather than multi-tenant isolation between organizations. These are reasonable simplifications for a lab-scope database project and are noted here so the scope boundary is explicit.

2. Entity-Relationship (ER) Diagram

The ER diagram below (created separately using an online diagramming tool) models the relational schema that backs RBEMS. It is derived directly from the tables, views, and stored procedures referenced throughout the PHP application layer. A short description of each entity and the relationships between them is given first so the diagram can be read against a clear key.

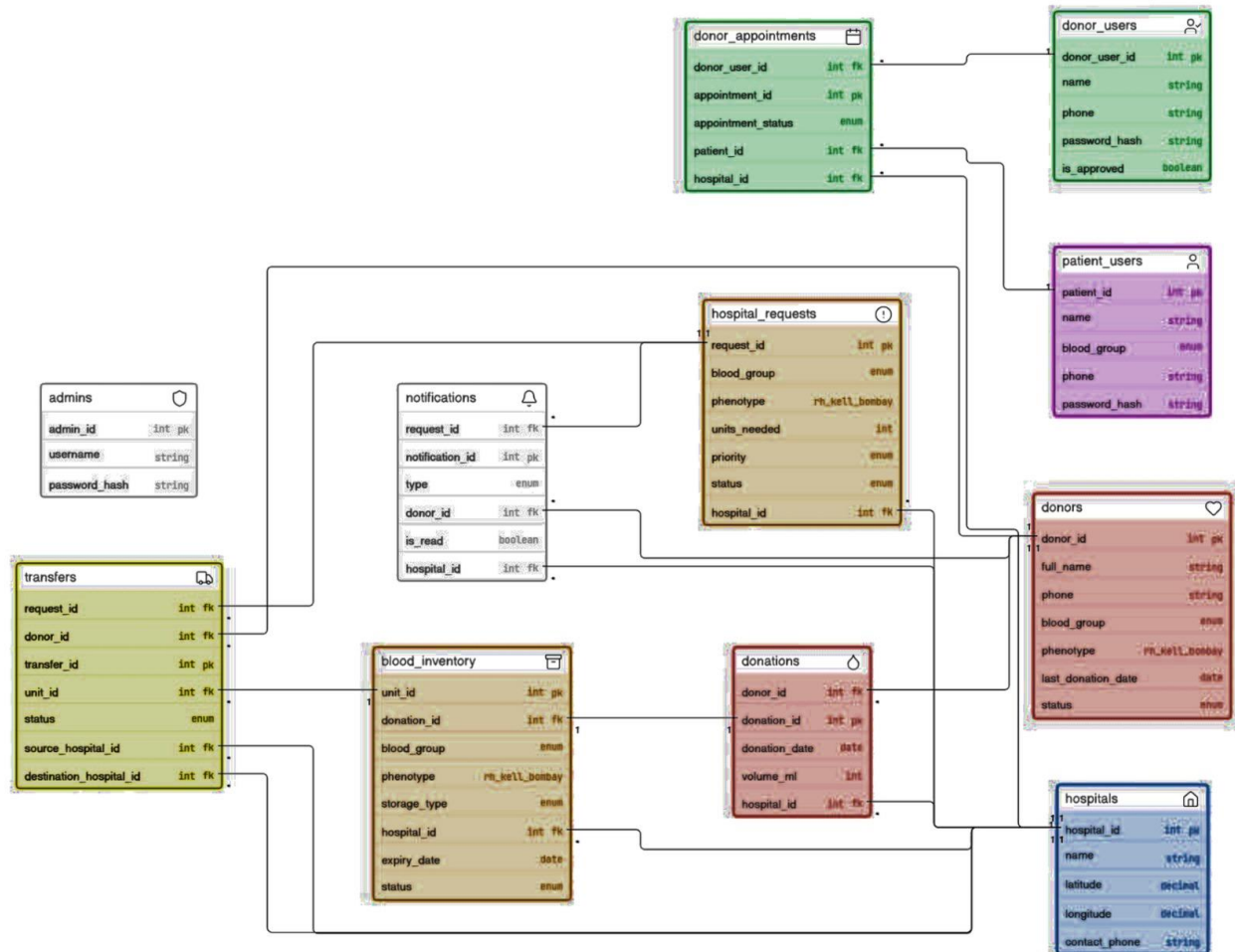


Fig : ER Diagram of our Rare Blood Emergency & Matching System

Core Entities

Entity (Table)	Purpose	Key Attributes
hospitals	Reference list of participating hospitals.	hospital_id (PK), name, address, contact_phone
admins	System administrator accounts.	admin_id (PK), username, password_hash
donors	Master donor record used by the compatibility engine (independent of donor login accounts).	donor_id (PK), full_name, phone, email, gender, dob, blood_group, rh_type, kell_status, bombay_phenotype, address, status, last_donation_date
donor_users	Donor self-service login accounts, subject to admin approval.	donor_user_id (PK), name, username, dob, address, phone, habits, password_hash, is_approved
patient_users	Patient self-service login accounts.	patient_id (PK), name, username, dob, weight, blood_group, address, phone, password_hash
blood_inventory	Physical stored units of blood at a hospital.	unit_id (PK), blood_group, rh_type, kell_status, bombay_phenotype, storage_type, collection_date, expiry_date (trigger-computed), hospital_id (FK), status
hospital_requests	A hospital's request for units of a given profile.	request_id (PK), hospital_id (FK), blood_group, rh_type, kell_status, bombay_phenotype, units_needed, priority, status, needed_by, request_date
transfers	Records the movement/allocation of a unit (or a mobilized donor) to satisfy a request.	transfer_id (PK), request_id (FK), unit_id (FK, nullable), source_hospital_id (FK, nullable), destination_hospital_id (FK), quantity, status, approved_by, transfer_date

Entity (Table)	Purpose	Key Attributes
donor_appointments	A confirmed donation meeting between a donor, a patient, and a hospital.	appointment_id (PK), donor_user_id (FK), hospital_id (FK), patient_id (FK), appointment_status, notes, created_at
notifications	Auto-logged system events (triggergenerated).	notification_id (PK), type, message, created_at

Derived Views and Stored Procedures

Three views and one stored procedure encapsulate business rules at the database layer so that every page reads consistent, pre-computed results rather than re-implementing the logic in PHP:

- v_eligible_donors — joins donors with their most recent donation date and applies the 90/120day interval rule to compute is_eligible and days_until_eligible.
- v_expiring_soon — filters blood_inventory to Available units expiring within 7 days.
- v_pending_requests — joins hospital_requests with hospitals for the main dashboard's pendingrequest panel.
- sp_find_compatible_inventory(group, rh, kell, bombay) — stored procedure used by compatibility_check.php to return every in-stock unit compatible with a given recipient profile, including the Rh-null/Bombay/Kell override rules.

Relationships Summary

- A hospital (1) can have many blood_inventory units, many hospital_requests, and can appear as either the source or destination of many transfers (0..N each).
- A hospital_requests row (1) can be linked to at most one transfers row that fulfils it (1..1 typical, enforced by application logic rather than a hard 1:1 constraint).
- A blood_inventory unit (0..1) may be referenced by one transfers row when it is reserved for that transfer.
- A donor_users account (1) can have many donor_appointments; each appointment also references exactly one patient_users row and one hospitals row.
- Triggers on hospital_requests and transfers each insert into notifications, so notifications has an implicit (unmodelled as an FK) 1:N logical source relationship with both tables.

3. System Outputs, Explanations, and Screenshots

This section walks through each of the eight implemented features, explains what the corresponding page does and which rule it enforces, and reserves space for the screenshot(s) that demonstrate it running against the seeded database.

Donor Registration and Registry

New donors are captured with their full phenotype profile through the registration form. Immediately afterwards, donor_list.php shows the full registry with each donor's computed eligibility (Eligible, or “In N day(s)”) pulled live from the v_eligible_donors view, and highlights rare attributes (Rh-null, Bombay, Kell+) with badges.

The screenshot shows a web form titled "Donor Registration" with the subtitle "Capture donor identity, contact details and rare phenotype profile." The form is divided into three main sections: Identity, Phenotype Profile, and a registration button.

Identity Section:

- Full Name:** Text input field containing "saka".
- Phone:** Text input field containing "6868".
- Email:** Text input field containing "saka@gmail.com".
- Gender:** Dropdown menu with "Male" selected.
- Date of Birth:** Text input field containing "11/10/2001".
- Address:** Text input field containing "dhanmondi".

Phenotype Profile Section:

- ABO Group:** Dropdown menu with "O- (uncommon)" selected.
- Rh Type:** Dropdown menu with "Negative" selected.
- Kell Status:** Dropdown menu with "Negative" selected.
- Bombay phenotype (very rare):** Checkbox that is checked.

Registration Button: A red button labeled "Register Donor".

Automatic Donation Eligibility

Eligibility is never entered manually. The `v_eligible_donors` view computes it from each donor's `last_donation_date` using the 90/120-day interval rule, so the registry and the compatibility search always agree on who can currently donate.

Donor Registry & Eligibility

Eligibility is computed automatically from donation history using the interval rule (90 days for male donors, 120 days for others).

ID	NAME	PHONE	PROFILE	LAST DONATION	ELIGIBILITY
#9	lamine yamal	1919	ON Bombay	Never	Eligible
#10	saka	6868	ON Bombay	Never	Eligible
#8	steven smith	014949	OP Bombay	Never	Eligible

Rare Blood Compatibility Cross-Matching

An admin or clinician enters the recipient's ABO group, Rh type, Kell status, and Bombay-phenotype flag. The page returns two ranked result sets: compatible units currently in stock and, as a fallback, compatible eligible donors (filtered in PHP with isCompatible() against the same rule set), so staff always see the nextbest option if no unit is available.

Rare Blood Compatibility Cross-Matching

Enter the recipient's phenotype to find compatible inventory units and eligible donors, including Rh-null, Bombay phenotype and Kell overrides.

Recipient ABO Group

Rh Type

Kell Status

O

Negative

Negative

☒ Bombay phenotype

Find Matches

Compatible Inventory Units

UNIT	TYPE	STORAGE	EXPIRES	HOSPITAL
#4	ABP Bombay	Cryopreserved	3391 day(s)	Evercare Hospital

Compatible Eligible Donors (fallback if stock unavailable)

ID	NAME	PHONE	PROFILE
#8	steven smith	014949	OP
#9	lamine yamal	1919	ON
#10	saka	6868	ON

Donor Registry & Eligibility

Eligibility is computed automatically from donation history using the interval rule (90 days for male donors, 120 days for others).

ID	NAME	PHONE	PROFILE	LAST DONATION	ELIGIBILITY
#9	lamine yamal	1919	ON Bombay	Never	Eligible
#10	saka	6868	ON Bombay	Never	Eligible
#8	steven smith	014949	OP Bombay	Never	Eligible

Inventory Shelf-Life and Expiry Alerts

Adding a unit only requires the collection date and storage type; the `trg_inventory_expiry` trigger computes the expiry date automatically (42 days for Fresh, 10 years for Cryopreserved). The inventory list sorts by status and expiry date and flags any available unit with 7 or fewer days remaining, matching the same units surfaced on the main dashboard's "Expiring Within 7 Days" panel.

Blood Inventory

Shelf-life is calculated automatically on entry (42 days fresh, 10 years cryopreserved) via a database trigger.

+ Add Unit

UNIT	TYPE	STORAGE	COLLECTED	EXPIRES	HOSPITAL	STATUS
#1	ON	Fresh	01 May 2026	12 Jun 2026 -88d left	Dhaka Medical College Hospital	Available
#5	AP	Fresh	01 Aug 2026	12 Sep 2026	Evercare Hospital	Available
#4	ABP 	Cryopreserved	01 Dec 2025	01 Dec 2035	Evercare Hospital	Available
#3	OR 	Cryopreserved	10 Jan 2026	10 Jan 2036	Dhaka Medical College Hospital	Available
#2	BN 	Cryopreserved	15 Jun 2026	15 Jun 2036	Square Hospital	Available

Hospital Request Submission with Priority Queueing

A hospital submits its need (profile, quantity, priority, needed-by time). Every listing of requests — on the dashboard, in the admin panel, and in the tracking page — orders Emergency before Urgent before Routine, so the most critical cases are always shown first regardless of submission order.

Request Status Tracking

All hospital requests, queued by priority and current status. Notifications are logged automatically on submission and fulfillment.

[+ New Request](#)

ID	HOSPITAL	NEED	PRIORITY	STATUS	NEEDED BY
#2	Evercare Hospital	AP × 2	Routine	Pending	21 Aug 2026, 02:50
#1	Square Hospital	DR × 1	Emergency	Approved	18 Aug 2026, 08:50

Admin Approval and Inter-Hospital Transfers

The admin dashboard is the control centre: pending requests can be approved or rejected; donor selfregistrations can be approved, rejected, or later deleted; and an approved request can be turned into a transfer, either against a specific reserved inventory unit or as a donor-mobilization case. Open transfers are tracked until marked Completed.

Login

Sign in as a patient, donor, or system admin.

Account Type

Admin

Username

admin1

Password

.....

Log In

Need a patient or donor account? [Create one here.](#)

Admin Dashboard

Signed in as admin1. Approve requests, then arrange inter-hospital unit transfers.

Requests Awaiting Action

ID	HOSPITAL	NEED	PRIORITY	STATUS	ACTIONS
#1	Square Hospital	OR × 1	Emergency	Approved	Ready for transfer below
#2	Evercare Hospital	AP × 2	Routine	Pending	<button>Approve</button> <button>Reject</button>

Donor Registrations Pending Approval

NAME	DOB	PHONE	HABITS	ACTION
Yeamin Chowdhury	2005-12-23	01540576816	Late Sleeping	<button>Approve</button> <button>Reject</button>
david	2004-03-02	01232323	walking	<button>Approve</button> <button>Reject</button>
smith	2003-06-12	01334343	Smoking	<button>Approve</button> <button>Reject</button>
Nabil Hasan	2006-12-14	0123456788	Gardening	<button>Approve</button> <button>Reject</button>
Samira Ahmed	2000-01-10	01710000012	Regular exercise and no recent illness	<button>Approve</button> <button>Reject</button>

Create Transfer / Fulfillment

Request	Inventory Unit (optional)	Source Hospital	Destination Hospital
#1 - Square Hospital (OR) ▾	— mobilize donor instead ▾	— ▾	Dhaka Medical College Hc ▾
Quantity			
1			
<button>Create Transfer</button>			

Open Transfers

ID	REQUEST	NEED	DESTINATION	STATUS	ACTION
#1	#1	OR	Evercare Hospital	Approved	<button>Mark Completed</button>

Request Status Tracking and Notifications

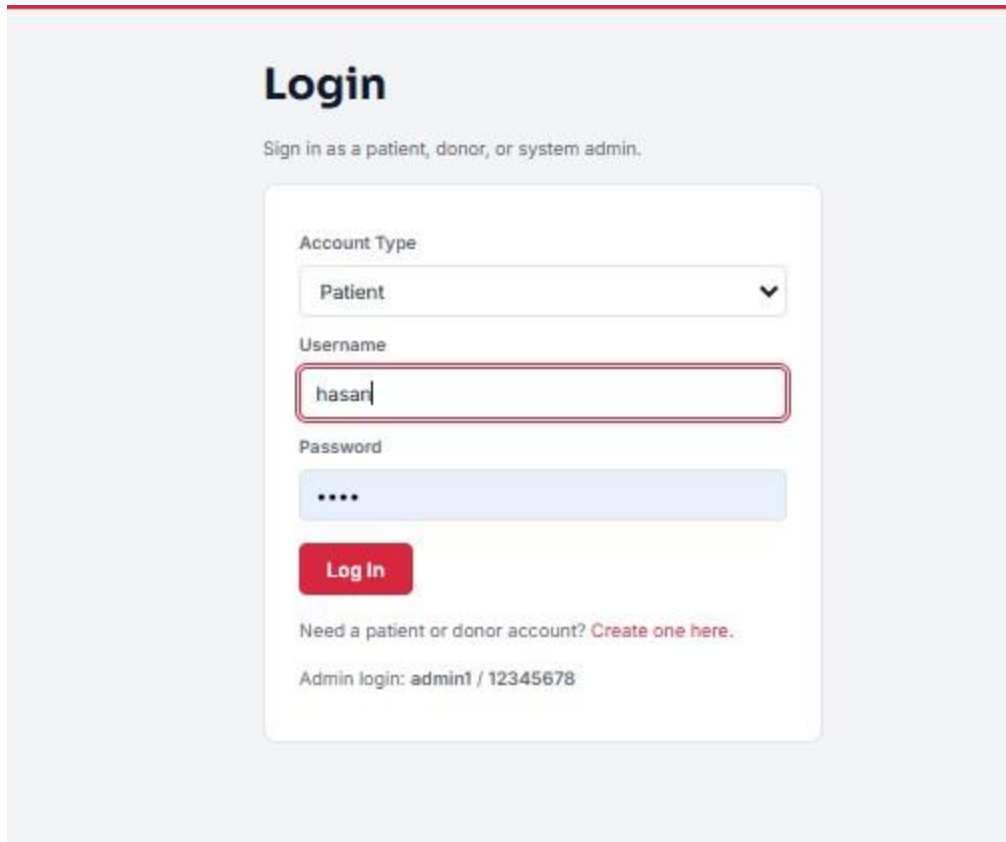
request_list.php gives hospitals a full history of every request (Pending, Approved, Fulfilled, Rejected) ordered by status then priority. notifications.php shows the automatically generated log produced by trg_request_notification (on submission) and trg_transfer_completed (on fulfilment) — no page inserts these rows manually, so the log cannot fall out of sync with the actual events.

Requests Awaiting Action					
ID	HOSPITAL	NEED	PRIORITY	STATUS	ACTIONS
#1	Square Hospital	OR × 1	Emergency	Approved	Ready for transfer below
#2	Evercare Hospital	AP × 2	Routine	Pending	<button>Approve</button> <button>Reject</button>

Notifications		
Auto-logged by database triggers whenever a request is submitted or a transfer completes.		
TYPE	MESSAGE	WHEN
Request	Emergency request #1 submitted for ORh-null (1 unit(s))	18 Aug 2026, 02:50
Request	Routine request #2 submitted for APositive (2 unit(s))	18 Aug 2026, 02:50

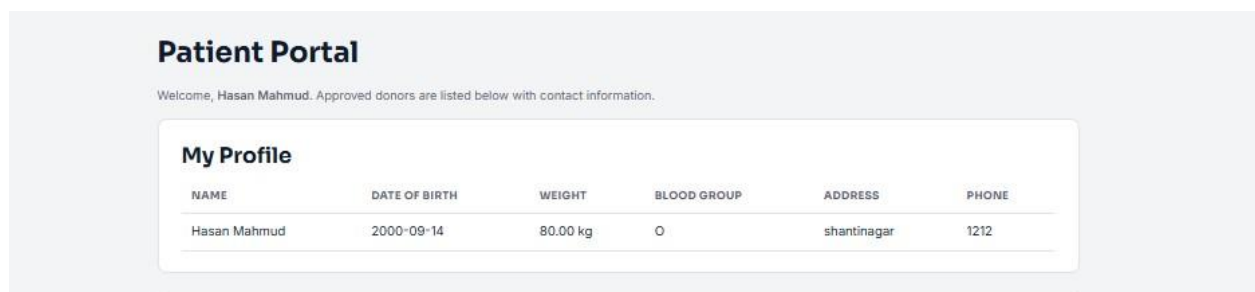
Role-Based Login and Dashboards

A single login page authenticates against three separate tables depending on the selected role. Successful login regenerates the session ID and clears any session state from a different role before redirecting to the matching dashboard (patient, donor, or admin), each of which is guarded by a `require*Login()` check.



The login page features a central white card on a light gray background. At the top of the card, the title 'Login' is displayed in a large, bold, dark font. Below the title, a subtitle reads 'Sign in as a patient, donor, or system admin.' The main form area contains three input fields: 'Account Type' with a dropdown menu currently showing 'Patient', 'Username' with the text 'hasan' entered, and 'Password' with four dots indicating masked input. A red 'Log In' button is positioned below the password field. At the bottom of the card, there is a link 'Need a patient or donor account? Create one here.' and a text entry 'Admin login: admin1 / 12345678'.

Fig : A simple login page which authenticates for 3 users (Patient, Donor and Admin)



The Patient Portal interface shows a header with the title 'Patient Portal' and a welcome message: 'Welcome, Hasan Mahmud. Approved donors are listed below with contact information.' Below this, a 'My Profile' section contains a table with the following data:

NAME	DATE OF BIRTH	WEIGHT	BLOOD GROUP	ADDRESS	PHONE
Hasan Mahmud	2000-09-14	80.00 kg	O	shantinagar	1212

Fig : Patient's Portal

Donor Portal

Welcome, Rodrygo. Search for a nearby hospital and confirm your donation meeting.

Your donation meeting has been confirmed.

Find and Confirm a Hospital

Search hospitals

Search by name, address, or contact

Search

Select Patient

Choose a patient

Select Hospital

Choose a hospital

Confirm Donation

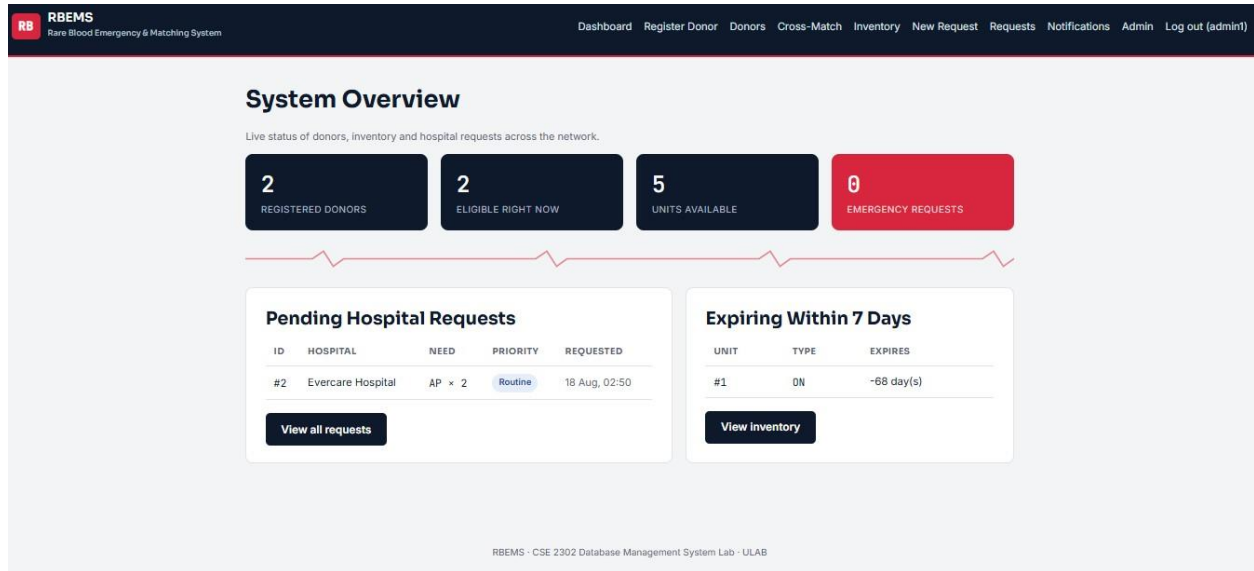
Your Confirmed Visits

PATIENT	HOSPITAL	STATUS	CONFIRMED AT
Hasan Mahmud	Dhaka Medical College Hospital	Confirmed	2026-08-20 03:16:36

Fig : Donor's Portal

System Overview Dashboard

index.php aggregates live counts (registered donors, currently eligible donors, available units, open emergency requests) alongside the pending-request queue and the expiring-soon list, giving a single at-a-glance operational summary that links out to the detailed pages.



Conclusions

The Rare Blood Emergency and Matching System (RBEMS) successfully demonstrates how a well-designed relational database can automate critical decision-making in a real-world healthcare scenario. By encoding rules such as donor eligibility, blood shelf-life, compatibility matching, and request prioritization directly into triggers, views, and stored procedures, the system ensures accuracy and consistency without relying on manual checks. The project covers all core database concepts — including multi-table relationships, prepared statements for security, and role-based access control — while solving a genuine problem: quickly locating compatible blood, especially for rare phenotypes like Rh-null and Bombay blood group. Overall, RBEMS shows how database automation can reduce human error and response time in emergency situations, making it a practical and meaningful application of DBMS concepts.